Sentence Style Detection for Author Switches

Author Switch Detection

Tshephang Phuti-A-Nkutu Matlala

Academy of Computer Science and Software Engineering
University of Johannesburg
South Africa



1 Problem Statement

2 Representation & Features

3 Models Overview

4 Results and Interpretation

5 Automated Ablation Study

6 Handling Class Imbalance

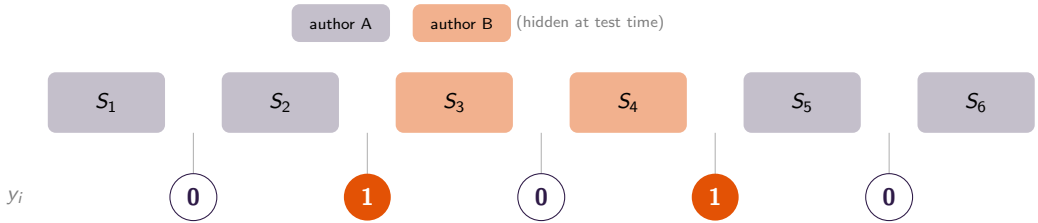
7 Future Work

8 Conclusion

9 Questions



Problem statement: binary classification



Formally: for a document $D = \{S_1, \ldots, S_n\}$, predict for every adjacent pair (S_i, S_{i+1}):

$$y_i = \begin{cases} 0 & S_i \text{ and } S_{i+1} \text{ share an author,} \\ 1 & \text{an author switch occurs between them.} \end{cases}$$

Why is it interesting?

1 **Difficult**

subtle style
topic controlled

2 **Short evidence**

one sentence
one small trace

3 **Descriptive cues**

length, punctuation,
vocabulary

4 **Interpretable**

we see which cues
drive a decision

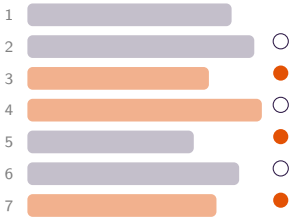
5 **Useful**

plagiarism,
co-authored text



Dataset: PAN 2026 multi-author writing style analysis

one problem = one document



problem-N.txt: one sentence per line
truth-problem-N.json: 0 / 1 per boundary

31,500
training problems
(10,500 per difficulty)

2.47M
sentences

2.44M
sentence pairs

14.3%
author switches

fanfiction-derived · easy / medium / hard

splits (by document)

Train
2,439,074 pairs

Validation
264,150 pairs

Test
259,653 pairs

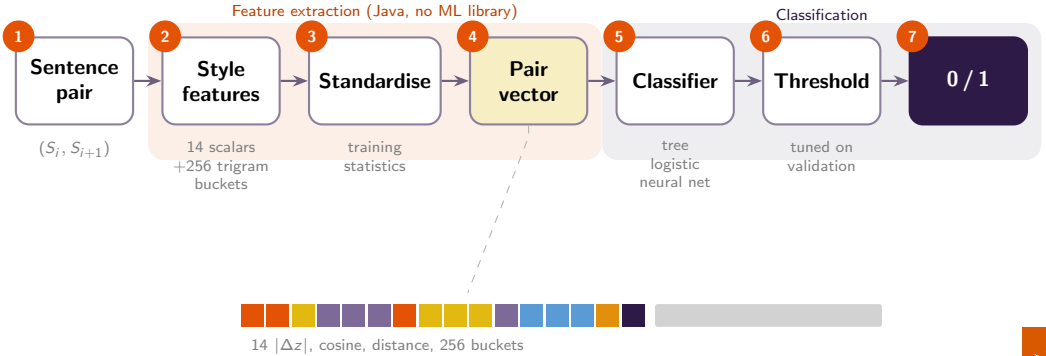
test **official validation, 2nd half** | halves **document disjoint, seed 42** | source **Zangerle et al., 2026**

Dataset: three difficulty levels

PAN 2026 · **2,439,074** training pairs · **14.3%** switches



Pipeline: from raw text to a prediction



Features: what the classifier sees

Length

words
characters
word length

Orthography

punctuation
digits
uppercase
punct. variety*

Vocabulary

frequent words
TF-IDF
emotion (NRC)
type-token*

Readability

Flesch ease*
Gunning Fog*
Flesch-Kincaid*

Character trigrams



hashed into 256 buckets

cosine similarity
+ 256 bucket differences

restored

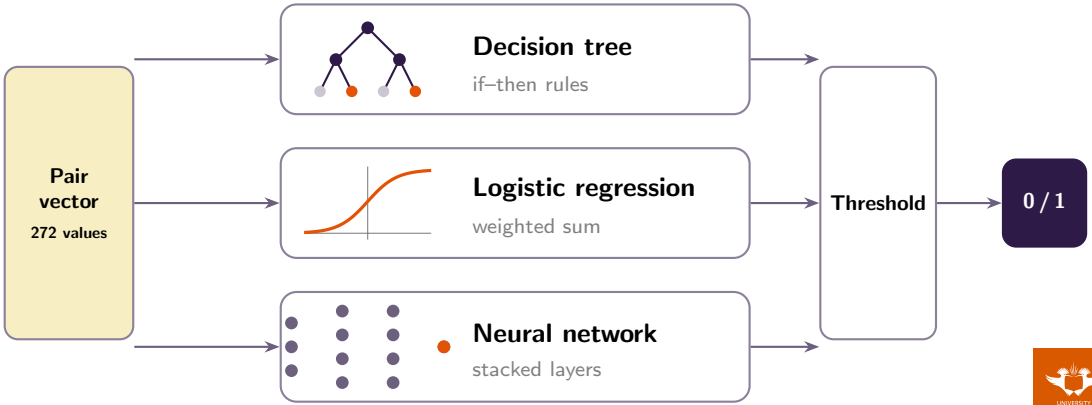
14 scalar features, z-scored with training statistics * added, more robust to topic

Pair vector

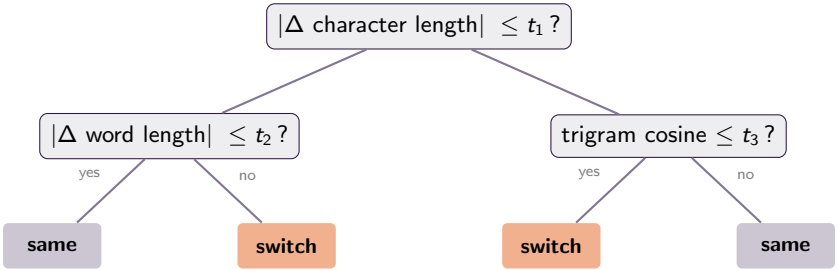


$14\,|\Delta z| + \text{cosine} + \text{distance} + 256$ bucket differences $=$ **272 values**

Three classifiers, one pair vector



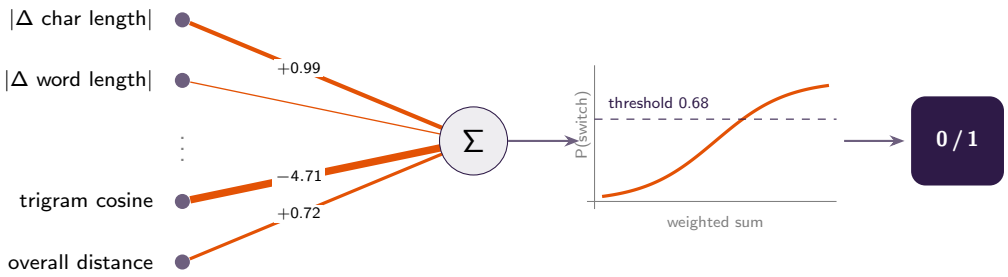
Model 1: Decision tree



criterion **Gini** | max depth **5** | min leaf **5** | class weighted **yes** | threshold **0.5**

Every prediction is a short, readable chain of rules.

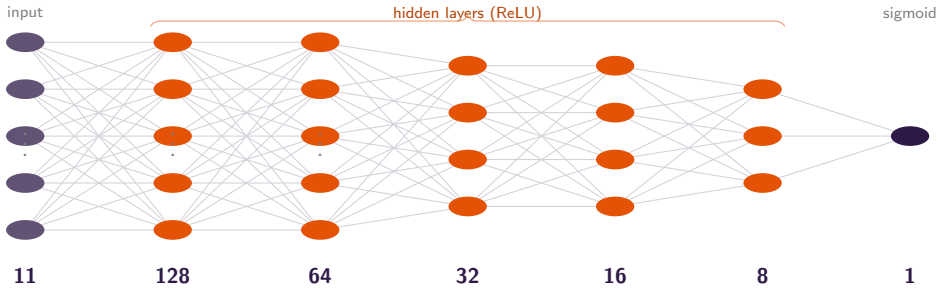
Model 2: Logistic regression



optimiser **batch gradient descent** | learning rate **0.6** | iterations **10,000** | class weighted **yes**

Line thickness = learned weight. The trigram cosine dominates.

Model 3: Neural network



training **mini-batch SGD** | learning rate **0.001** | dropout **0.2** | L2 **0.01** | early stopping **yes**

Most flexible model, yet not better than the linear one (next slides).

Results: overall held out test (259,653 pairs)

Model	Acc.	Prec.	Rec.	F1	Bal. Acc.
Decision tree	0.850	0.462	**0.410**	0.435	0.666
Logistic regression	0.869	0.546	0.392	**0.456**	**0.669**
Neural network	**0.871**	**0.562**	0.368	0.445	0.661

majority baseline accuracy **0.859**

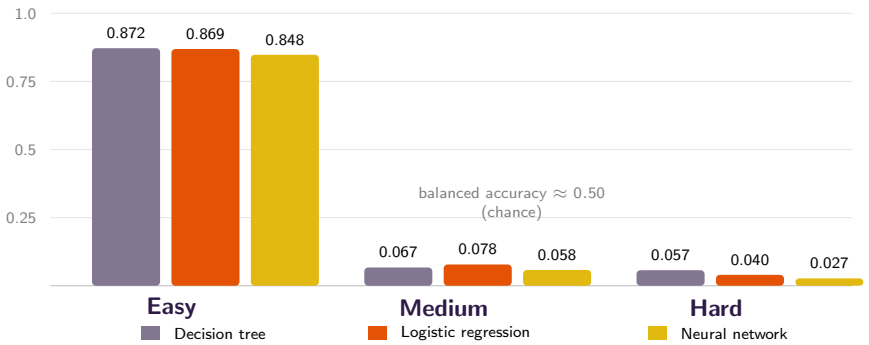
best F1 **logistic regression**

models **nearly tied**

first run **9 style features, 11-value pair vector**



Results by difficulty (F1)



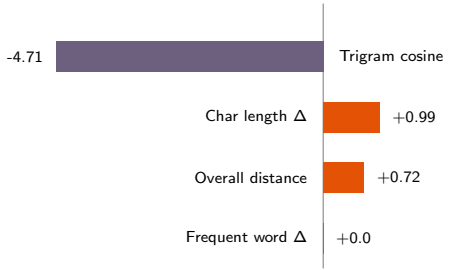
Interpretation: what the models used

Decision tree: splits used

1. Root: char length Δ
2. Then: word length Δ
3. Trigram cosine (almost every level)

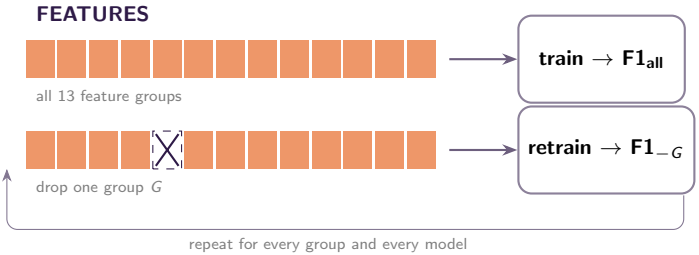
only **5 of 11** features used

Logistic regression: weights

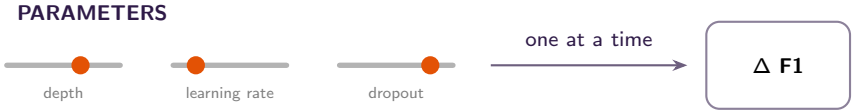


reading **mostly surface statistics** | model **first run, 11 features**

Ablation study: automated

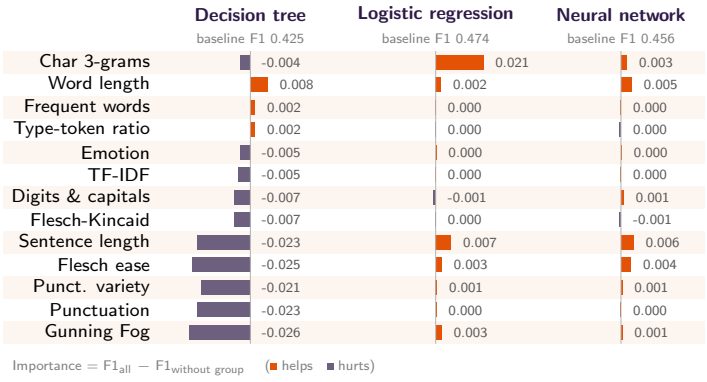


Importance(G)
$F1_{all} - F1_{-G}$



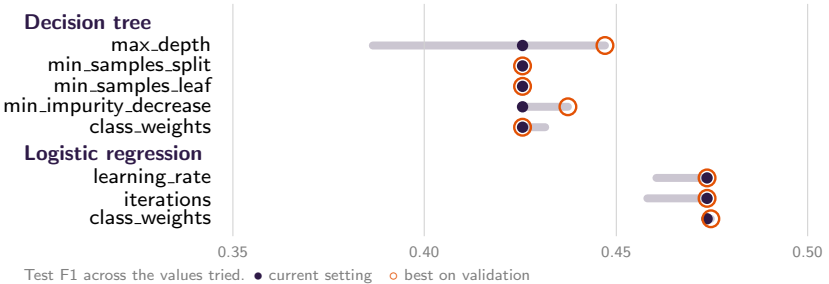
models **3** | feature groups **13** | parameter sweeps **15** | output **CSV, Markdown, LaTeX**

Ablation: which feature groups matter?



runs **1 seed, held out test split** features **16-value pair vector**

Ablation: which parameters matter?



status **partial: neural network parameter sweeps still running**

Handling class imbalance



"always same author" already looks accurate

1. Class weights

same 0.583 switch 3.505

2. Tuned threshold

maximises F1 on validation

3. Fair metrics

balanced accuracy, F1, majority baseline

resampling **none, all 2.4M pairs kept**

Future work

1

Retrain and re-evaluate

new 14-feature set
with trigram buckets

2

POS features

tagger not yet
implemented

3

Stronger models

random forest,
gradient boosting

Summary

272

pair features:
14 styles + cosine + distance + 256 buckets

0.44–0.46

baseline F1
of the three models

3 + 1

classifiers and an automated ablation study

Thank You

Questions?

